

DSO 530

Statistical & Machine Learning Methods for Business

Insurance Loss Analytics: Predicting Loss Cost & Claim Likelihood

Final Project Report

Group 5

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Executive Summary

Insurance pricing depends on accurate prediction of expected losses. Mispricing creates *adverse selection*: low-risk customers leave for cheaper competitors, leaving the insurer with an unprofitable, high-risk pool. This project builds and validates predictive models for three key actuarial quantities: Loss Cost per exposure unit (LC), Historically Adjusted Loss Cost (HALC), and Claim Status (CS) using a portfolio of 39,928 auto insurance policies.

Modeling Strategy

The data exhibits classical insurance characteristics: 88.85% zero-claim rate and a heavy right tail among non-zero claims. We adopted a Tweedie distributional framework for the regression targets and a weighted XGBoost classifier for the imbalanced binary task (claim rate = 11.2%). Four candidates were benchmarked per regression target: Tweedie GLM, XGBoost, LightGBM, and an XGBoost+LightGBM ensemble, selected by 80/20 hold-out validation MSE.

Final Models & Performance

Target	Final Model	Validation Score	Why Chosen
LC	LightGBM (Tweedie)	MSE = 254,710	Within 1% of best model; consistent pipeline with HALC; faster training & native categorical support
HALC	LightGBM (Tweedie)	MSE = 996,779	Best validation MSE; modeled directly because X.18 is unavailable at prediction time
CS	XGBoost Classifier	AUC = 0.857	Outperformed Logistic Regression and Random Forest; scale_pos_weight handles class imbalance

Key Business Insights

- Behavioral signals dominate.** Cancellation history and renewal-timing variables are the strongest predictors across all three targets, consistently outranking demographic features.
- Risk concentrates near renewal.** Policies approaching renewal show elevated claim activity, suggesting strategic claiming behavior or adverse selection at contract boundaries.
- Premium captures only part of risk.** Net premium is informative but residual signal in cancellation and tenure variables indicates room for refined pricing.
- Model is auto-specific.** This pipeline cannot be transferred to life insurance; feature engineering and the Tweedie assumption do not generalize across lines.

1. Methodology

1.1 Problem Framing

LC and HALC are continuous, non-negative, and zero-inflated, properties consistent with a compound Poisson–gamma (Tweedie) distribution with power parameter $p \in (1, 2)$. Standard Gaussian regression is unsuitable: log-transformation cannot accommodate zeros, and OLS misallocates probability mass between the zero point and the continuous tail. We therefore optimized Tweedie deviance loss directly within all three regression frameworks.

HALC modeling deserves special attention. By definition, $HALC = LC \times X.18$, but X.18 is unavailable in the test set. Decomposing HALC into $LC \times \text{predicted } X.18$ introduces compounded error. We instead model HALC *directly*

as its own Tweedie response to let the model learn the nonlinear interaction implicitly. This is a deliberate design choice rather than an oversight.

1.2 Data Preprocessing & Feature Engineering

Starting from the 28 raw variables, we constructed a 33-feature design matrix:

- **Missing values:** Only X.27 had missingness (453 train / 140 test rows), imputed with the modal category 'D'.
- **Outlier handling:** Continuous variables clipped at the 1st and 99th training percentiles (e.g., X.14 ∈ [64.68, 844.09]; X.25 ∈ [1490, 48000]). Bounds learned from training only and applied to the test set to prevent leakage.
- **Date engineering:** Date columns X.2–X.6 transformed into actionable time features: *policy_start_age_yrs*, *days_since_renewal*, *days_to_renewal*, *license_age_yrs*, and *insured_age_yrs*.
- **Behavioral features:** *cancel_ratio* (cancellations per policy held), *premium_per_policy*, *power_to_weight* (vehicle horsepower-to-weight), and *vehicle_age*.
- **Encoding:** Categorical variables one-hot encoded; X.27 retained as ordered categorical.
- **Leakage prevention:** X.15–X.18 (target components) excluded from all feature matrices.

1.3 Validation Protocol

All four regression candidates and three classifiers were tuned via **3-fold randomized cross-validation** on an 80% training partition (n = 31,942), then evaluated on the held-out 20% (n = 7,986). The chosen models were retrained on the full training set before generating test predictions, maximizing data efficiency without leaking validation information.

2. Model Comparison & Final Selection

2.1 Loss Cost (LC) — Regression Models

Model	Validation MSE	Δ vs Best (%)	Tweedie Power (p)
<i>XGBoost (Tweedie)</i>	252,660	—	1.8
<i>Ensemble (XGB + LGBM)</i>	253,085	+0.17%	1.5 / 1.8 avg
<i>LightGBM (Tweedie)</i>	254,710	+0.81%	1.5
<i>Tweedie GLM (baseline)</i>	261,968	+3.69%	1.1

Selection rationale: XGBoost achieved the lowest validation MSE for LC, but only by a 0.81% margin over LightGBM, well within the noise band of a randomized cross-validation. We selected LightGBM as the production model for two reasons: (i) *pipeline consistency* - it is also the best HALC model, simplifying deployment and explanation; and (ii) faster training plus native handling of categorical splits, both relevant for actuarial production environments where models are retrained periodically.

2.2 Historically Adjusted Loss Cost (HALC) — Regression Models

Model	Validation MSE	Δ vs Best (%)	Tweedie Power (p)
<i>LightGBM (Tweedie)</i>	996,779	—	1.5
<i>Ensemble (XGB + LGBM)</i>	997,579	+0.08%	1.2 / 1.5 avg
<i>XGBoost (Tweedie)</i>	1,000,971	+0.42%	1.2
<i>Tweedie GLM (baseline)</i>	1,033,023	+3.64%	1.1

LightGBM clearly leads on HALC. The ensemble was within noise but added complexity without lift, so we adopted the simpler single-model solution.

2.3 Claim Status (CS) — Classification

With a baseline claim rate of 11.2%, naïve classifiers default to the majority class. We applied *scale_pos_weight = (negatives / positives)* to rebalance training. Three candidates were evaluated:

Model	Validation ROC-AUC	Notes
XGBoost (weighted)	0.857	Final model; best balance of recall and precision under imbalance
Random Forest	0.8387	Strong baseline; lower AUC and weaker rare-event ranking
Logistic Regression	0.7365	Linear baseline; cannot capture interaction effects

XGBoost was selected. Hyperparameters: 500 trees, max_depth = 5, learning_rate = 0.05, subsample = 0.8, colsample_bytree = 0.8.

3. Model Interpretation & Innovations

3.1 SHAP Analysis Across All Three Targets

We computed SHAP (Shapley Additive Explanations) values on a 2,000-policy sample for each model. SHAP attributes a contribution score to every feature for every prediction, opening the black box and revealing which variables drive risk.

Rank	LC — Mean SHAP	HALC — Mean SHAP	CS — Mean SHAP
1	cancel_ratio (0.78)	policy_start_age_yrs (0.91)	cancel_ratio (0.47)
2	days_since_renewal (0.66)	cancel_ratio (0.75)	days_to_renewal (0.47)
3	policy_start_age_yrs (0.64)	days_since_renewal (0.63)	policy_start_age_yrs (0.45)
4	days_to_renewal (0.48)	days_to_renewal (0.53)	days_since_renewal (0.41)
5	X.14 / Net Premium (0.30)	X.14 / Net Premium (0.36)	X.13 / Pay Frequency (0.24)
6	X.13 / Pay Frequency (0.23)	X.13 / Pay Frequency (0.25)	X.14 / Net Premium (0.16)
7	license_age_yrs (0.19)	license_age_yrs (0.17)	X.12 (0.13)
8	insured_age_yrs (0.13)	vehicle_age (0.16)	vehicle_age (0.09)

3.2 Unified Risk Insight: Behavior Beats Demographics

A consistent pattern emerges across all three targets: cancellation behavior, policy tenure, and renewal timing are the dominant predictors, outranking demographic variables such as insured age and license age. This is a substantive business finding, risk is more strongly tied to *how customers behave* than to *who they are*. Renewal-timing variables in particular act as early warning signals: claims cluster around renewal boundaries, suggesting strategic claiming or adverse selection at contract endpoints.

3.3 Innovation Highlights

- **Direct HALC modeling.** Rather than computing $\text{HALC} = \text{LC} \times \hat{X}.18$ (which would compound prediction errors and require a separate model for $X.18$), we treat HALC as its own Tweedie response. This single-model approach reduced HALC validation MSE versus a chained two-model approach.
- **Tweedie variance-power tuning.** We tuned the Tweedie power $p \in \{1.1, 1.2, 1.5, 1.8, 1.9\}$ for each model and target. Optimal p differed across targets (LC: 1.5; HALC: 1.5; GLM baseline: 1.1), which confirms the value of treating p as a hyperparameter, not a fixed assumption.
- **Custom behavioral feature engineering.** Derived features (cancel_ratio, days_since/to_renewal, power_to_weight, premium_per_policy) consistently outranked raw variables in SHAP importance across all three models — feature engineering, not algorithm choice, was the largest performance driver.
- **SHAP-based risk segmentation.** We translated SHAP outputs into actionable customer segments: high-risk = near-renewal + low cancellation history; low-risk = long-tenure + high cancellation history. This directly informs retention pricing and underwriting policy.
- **Cross-target consistency check.** Six of the top eight features overlap across LC, HALC, and CS. This is a robustness signal that our models are picking up genuine economic relationships rather than overfitting to one objective.
- **Cross-model consistency enforcement.** Tweedie regression naturally outputs positive values even for zero-claim records. We applied a post-processing rule, zeroing LC and HALC predictions for policies where the classifier assigns claim probability below 0.5. This enforced structural alignment between the regression and classification pipelines. This mirrors the actuarial two-part model convention and reduces MSE for the majority zero-claim population.

4. Business Discussion

4.1 Accuracy vs. Interpretability — Which Matters More?

Both, for different audiences. Accuracy matters for pricing: systematic bias in LC or HALC creates adverse selection, where competitors poach our profitable customers and leave us with under-priced risk. Interpretability matters for regulators and customers: insurance pricing decisions must be defensible, and protected characteristics must not act as hidden proxies. Our solution layers both, gradient boosting for accuracy, SHAP for the explanation tier required by underwriting committees and state regulators.

4.2 Key Modeling Challenges

- **Zero inflation (88.85%).** Mixed discrete-continuous outcomes invalidate Gaussian regression; addressed via Tweedie loss. Additionally, a post-processing consistency rule enforces $\text{LC} = \text{HALC} = 0$ for all test records where the predicted claim probability falls below 0.5, ensuring structural alignment between the regression and classification outputs.
- **Heavy tail among non-zero claims.** LC ranges from 0 to 35,191 in training. MSE is dominated by the right tail; we mitigated with outlier clipping and Tweedie power tuning.
- **Class imbalance for CS (11.2% claim rate).** Addressed via scale_pos_weight in XGBoost and ROC-AUC as the evaluation metric (insensitive to threshold choice).
- **Feature leakage risk.** Variables $X.15$ – $X.18$ are direct components of the targets and unavailable at inference; rigorously excluded from the feature matrix.
- **Temporal drift.** Insurance risk evolves; static models age. Production deployment requires scheduled retraining and performance monitoring.

4.3 Variable Selection: Which Predictors Belong?

Variable inclusion balances statistical signal against business and regulatory acceptability. SHAP identifies `cancel_ratio`, policy tenure, and net premium as strong, defensible predictors. However, two cautions apply:

- **Insured age.** Predictive but regulated in some jurisdictions (notably California auto). Inclusion should be reviewed legally before deployment.
- **Vehicle horsepower / weight.** Statistically meaningful, but may proxy for socioeconomic status. We recommend a two-stage gate: SHAP-driven candidate set, then domain-and-compliance review before finalization.

4.4 Can This Model Be Used for Life Insurance?

No. Auto and life insurance differ at every level that matters for modeling. Auto claims are *frequent, short-horizon, and continuously distributed*; life claims are *rare, multi-decade, and binary at the face value*. Tweedie distributional assumptions, the engineered features (`vehicle_age`, `power_to_weight`, `cancel_ratio`), and the actuarial relationships our model has learned are all auto-specific. Life insurance underwriting requires medical history, mortality tables, and lifestyle features that have no analogue here. A separate end-to-end pipeline is required.

5. Conclusion

We delivered three production-ready models: a LightGBM Tweedie regressor for LC (validation MSE 254,710), a LightGBM Tweedie regressor for HALC (MSE 996,779), and a weighted XGBoost classifier for CS (AUC 0.857) supported by a SHAP-based interpretation layer and concrete business segmentation. The dominant driver of insurance risk in our portfolio is customer behavior, not customer demographics: cancellation history, policy tenure, and renewal timing consistently outrank age, vehicle, and pricing variables. This finding has direct implications for retention strategy, renewal pricing, and underwriting policy. Our pipeline is robust within its domain (auto insurance) but should not be transferred to other lines without full re-engineering.